Grant Parker

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Summary: Self-motivated computer science student with software engineering experience including artificial intelligence and machine learning, industry standard programming languages, and software revision management.

Education: B.S. Computer Science, CU Boulder, Dept GPA 3.61 (Expected December 2026)
Westlake High School - Austin, TX (Cum Laude)

Experience and Internships:

Intern Software Engineer - Festo Corporation - IM Automation and Emerging Technologies
Agentic AI-Driven Frontend Generator (In Progress) - (August 2025 - May 2026)

Developing an intelligent, agent-based system for Festo, a global automation company headquartered in Germany, that automatically generates production ready frontend interfaces aligned with DevOps standards. Applies AI driven intent recognition, orchestrator, NLP based language learning model, RAG, and Claude via Anthropic API to interpret user stories and mockups into production-ready standards-compliant code. Designed as an AI assisted code generation pipeline, this project streamlines development and accelerates delivery across enterprise applications. Working in Agile sprints organized by product epics and user stories, collaborating with an international team on system architecture, DevOps integration, and user experience alignment.

Undergraduate Researcher - University of Colorado Boulder, Department of Computer Science - PhyPID Group
Software Development Research - (Summer 2024)

PhyPID research group member in the CU Boulder department of computer science responsible for integrating Ratel, a GPU optimized next generation solid mechanics solver into FreeCAD, an open-source computer aided design software. Added capabilities of invoking Ratel through the FreeCAD GUI, and built unit test infrastructure to ensure code integrity. Additionally handled data translation between FreeCAD and Ratel. Languages used for this role were C++ and Python.

Projects:

High Frequency Market Making Agent via Reinforcement Learning
Machine Learning Engineer - (August 2025 – December 2025)

Built a reinforcement learning market-making SARSA (State-Action-Reward-State-Action) model on real order book and trade data, implementing a custom limit order book replay simulator that aligns millisecond snapshots with executed trades. Trained an on-policy agent with linear function approximation and tile coding over a 9-feature state (inventory, quote offsets, spread, midprice change, volume imbalance, signed trade volume, volatility, RSI) and a discrete action space. Evaluated against a fixed quoting baseline.

Skills and Certifications: Git, C++, Python, C, Java, Swift, Handlebars, JavaScript, NodeJS, HTML, CSS, Scala, SQL, Figma, Solidworks 3D CAD Certified, MATLAB, COMSOL, LaTeX, AWS, OOP, UML, Design Patterns, LLM, AGI

Relevant Coursework: Machine learning, AI, Object-oriented analysis and design, Software development methods and tools, Database systems, Data structures, Algorithms, Computer and CPU systems, Linear algebra, Discrete mathematics, Calculus series, Physics, Chemistry, Engineering design intent, Project architecture, Graphic design, Functional programming, Cybersecurity, Conflict management, Neural Networks, Deep Learning, Natural Language Processing

Leadership and Service Activities:

- Section lead for low brass in the Westlake marching band – managed section audits for ~30 band members
- Service animal raiser, Guide Dogs for the Blind

Hobbies: Jazz Trombone, Hiking, Skiing, Golf, Building desktop computers, Building computer keyboards, Sneakers

Reference available upon request. Open to relocation. US Citizen.